EXP NO: 06
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Hamming Code - Error detection and Correction

Aim:

To write hamming codes for error detection and correction.

code:

```
def hamming_encryption(data):
    k = len(data)
    r = 0
    while (2**r) <= (k+r+1):
        r += 1
    r -= 1
    n = k+r+1
    code = [0]*n
    j = 0
    for i in range(1,n):
        if (i & (i-1)) != 0:
            code[i] = data[j]
            j += 1
    for j in range(1,n)
        if j & pos:
        pair n = int [code[j]
    code.[pos] = str(pair)
    return (code[1:])
```

```
def hamming code - decryption (data, r, k):
for i in range (r):
    pos = 2++ :
    pair = 0
    for j in range (1, k + k + 1):
        if j < pos:
            pair x = int (code [j])
    if (pair != 0)
        if (data [pos ] = = 0):
            data [pos] = '1'

        else:
            data [pos] = '0'
    return : join (data)
    data = input ("Enter binary data:"
    print ("Data after encoding from
        sender side: ", hamming -
            encryption)
    rdata = Input ("enter data received
        by the reciever")
    print ( hamming code - decryption
        (rdata , r, rdatalength))
```

O/P:-

Enter binary data = 1011

Data after encoding from senders side = 0110011

Enter data received by = 0110111

Data after decoding : 0110011

Result:

Thus, the hamming code was detected and corrected the error successfully.